Ye Zhenpeng

Male 22 |+86 13724490933 |3115325549@qq.com Language：Chinese、Cantonese、English

My Profile：https://github.com/ZephyrZP

Professional Profile

Shaoguan University | Teaching Assistant (Neural Network & Deep Learning)

Sep 2025 - Nov 2026

Assisted in undergraduate deep learning tutorials, guiding students on model training and project development, marking assignments and debugging codes.

Guangzhou Baiyin Yun Biotechnology Co., Ltd.

Algorithm Engineer Dec 2025 - Mar 2026

TxAgent Medical Diagnostic Assistant Vertical LLM Agent **Deep Learning** Dec 2025 - Jan 2026

Tech Stack: Qwen2-7B, LoRA/PEFT Fine-tuning, RAG, LangGraph, Function Calling, Clinical Knowledge Base

• Designed multi-node controllable Agent workflow via LangGraph, modularised consultation, retrieval, diagnosis and plan generation, enabling traceable diagnosis logic and multi-turn context managemen

- Built structured clinical guideline knowledge base with RAG to align LLM outputs with clinical protocols; applied LoRA fine-tuning on medical corpus to improve response factual accuracy
- Wrapped lab test interpretation and drug contraindication tools, implemented Function Calling for autonomous tool invocation, supporting multi-department clinical decision assistance

Medical Image Diagnostic Prediction Model

Tech Stack: PyTorch, UNet++, TransUNet, Multi-scale Attention, Medical Data Augmentation, Mixed Precision Training, Dice Loss / Focal Loss

- Optimised UNet++ backbone with multi-scale attention and residual connections to enhance tiny lesion feature extraction, improved segmentation Dice coefficient by 12% over baseline
- Adopted medical-specific augmentation with 5-fold cross-validation and mixed precision training, mitigated overfitting from limited data and boosted model generalisation
- Developed multi-task pipeline for lesion segmentation and benign/malignant classification, output diagnosis confidence and quantitative metrics, achieved AUC=0.92 on multi-site imaging test set

Lightweight ST-GCN Real-time Human Action Recognition System 2026.6-2026.11

Tech Stack: PyTorch, MediaPipe Pose, ST-GCN, CA-AGC Adaptive Graph Convolution, Leave-One-Video-Out Cross Validation

- Adopted a skeleton-driven action recognition scheme, extracting 33 human keypoints via MediaPipe; designed the CA-AGC confidence-aware adaptive graph convolution module to dynamically adjust graph structure and optimize spatial feature aggregation, improving model robustness under occlusion and motion blur
- Built a lightweight spatial-temporal graph convolution network with only 0.41M parameters and 2ms per-sample inference, enabling real-time action recognition for webcam and video streams on low-end GPU RTX 3050
- Covered the full pipeline of data preprocessing, model training, leave-one-video-out cross validation, visual evaluation and real-time demo; supported zero-code addition of new action categories with good scalability

Academic Research & Publication

Oct 2024 - May 2025 | Medical NLP Research

Research Paper: LSTM-CRF based Clinical Named Entity Recognition for Electronic Medical Records

Tech Stack: Python, Keras, Bi-LSTM, CRF, Word2Vec/GloVe, BIO Sequence Labelling, Chinese Medical Text NER

- Devised entity-preserving text preprocessing pipeline and BIO annotation schema to eliminate error propagation from improper segmentation of professional Chinese medical terminology, lifting entity boundary recognition accuracy.
- Constructed Bi-LSTM-CRF hybrid model combining long-distance context modelling of LSTM and sequence label constraint of CRF to overcome performance limitations of single-structure models.
- Integrated pre-trained word embedding, Dropout regularisation and consistency verification layer to reduce overfitting; reached F1=93.4% on clinical EMR dataset, outperforming traditional CRF and standalone LSTM baselines.

Education Background

Shaoguan University | Bachelor of Electronic Information Science and Technology
Sep 2023 - Jun 2027 (Expected Graduation)

Technical Skills

Language/IDE：Python、shell、Vscode、cursor、Trae、Matlab Frameworks：Pytorch、Tensorflows、Keras、Opencv
Domain Knowledge：Windows、Linux 、Deep Learning、NLP、computer vision CI/CD：Github、Git、CSDN 、Docer

Awards & Competitions

1. **National First Prize, National College Students Engineering Practice & Innovation Competition (2025)**
2. **National Third Prize, Ruikang Robot Developer Competition (2025)**
3. **Provincial First Prize, National College Students Engineering Practice & Innovation Competition (2025)**
4. **Provincial First Prize, Ruikang Robot Developer Competition (2025)**
5. **Provincial Third Prize, Guangdong Engineering Practice & Innovation Competition (2024)**
6. **Individual Research Innovation Award, Shaoguan University (2024–2025)**